

printf():

It is a predefined function available in stdio.h

It is the major output function in c language.

Printf always refer standard output device i.e. monitor.

In printf, **f means formatted**.

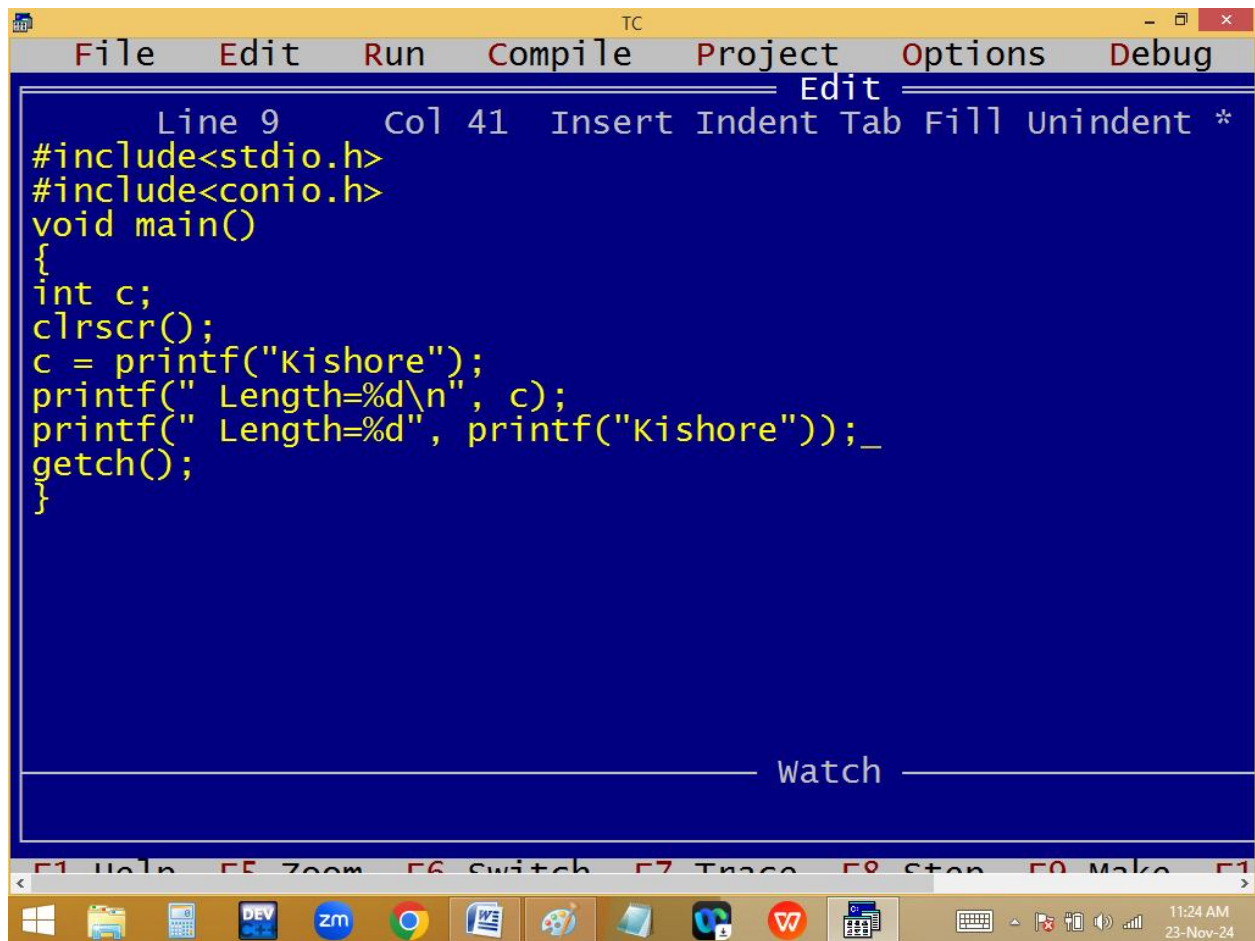
Syntax:

```
int printf("[text] [ conversion characters / format specifiers /  
format string ] " [ , variables ] [ , expressions ] );
```

Note:

1. printf always return int that indicates the no of visible characters in " ".
2. In printf execution order is right to left but printing is left to right.
3. In printf everything printed as it is except back slash characters and format specifiers.
4. Printf can perform both formatted unformatted outputs.
5. In printf the first argument should be in " "

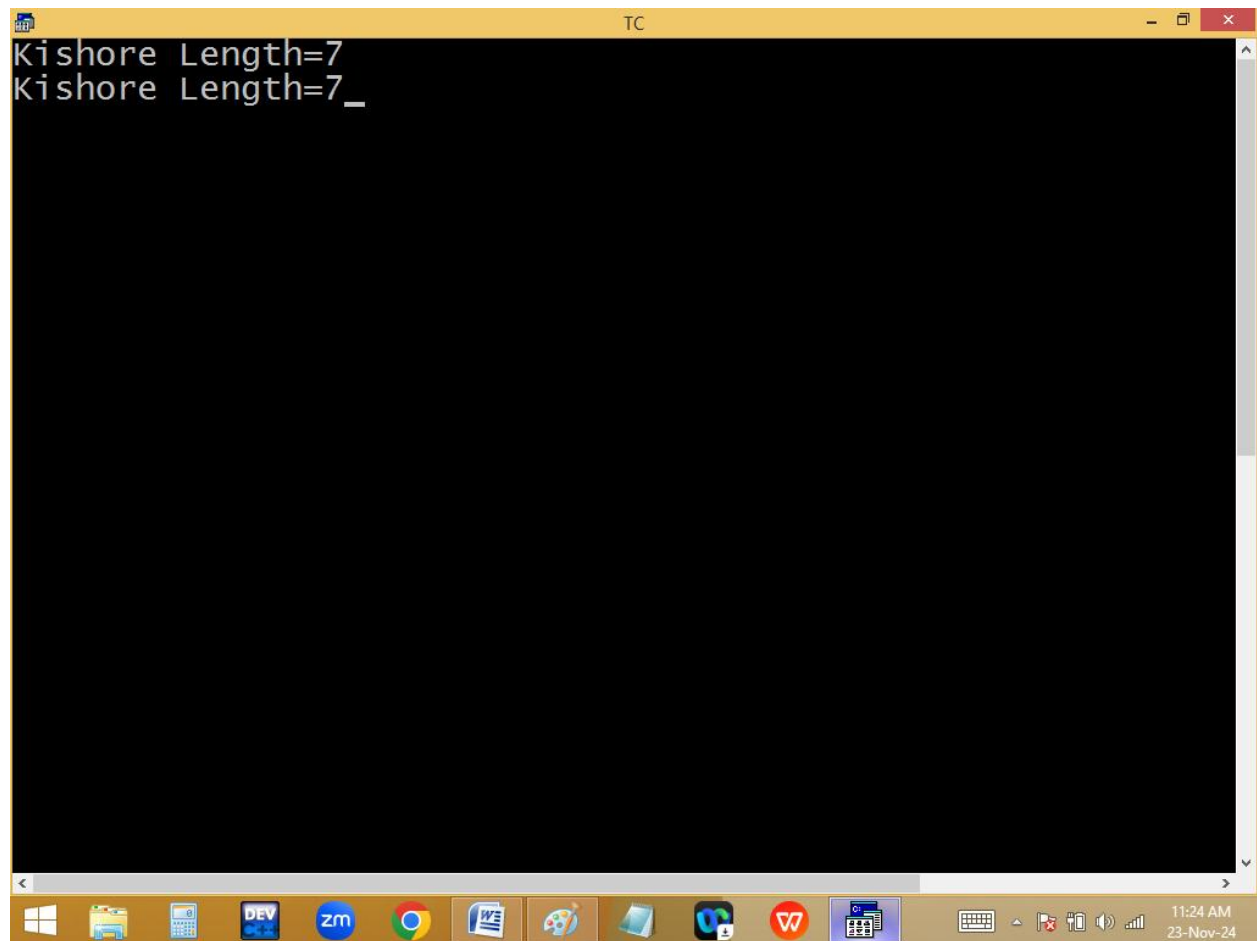
write a c program to find the string length without using loop / strlen().

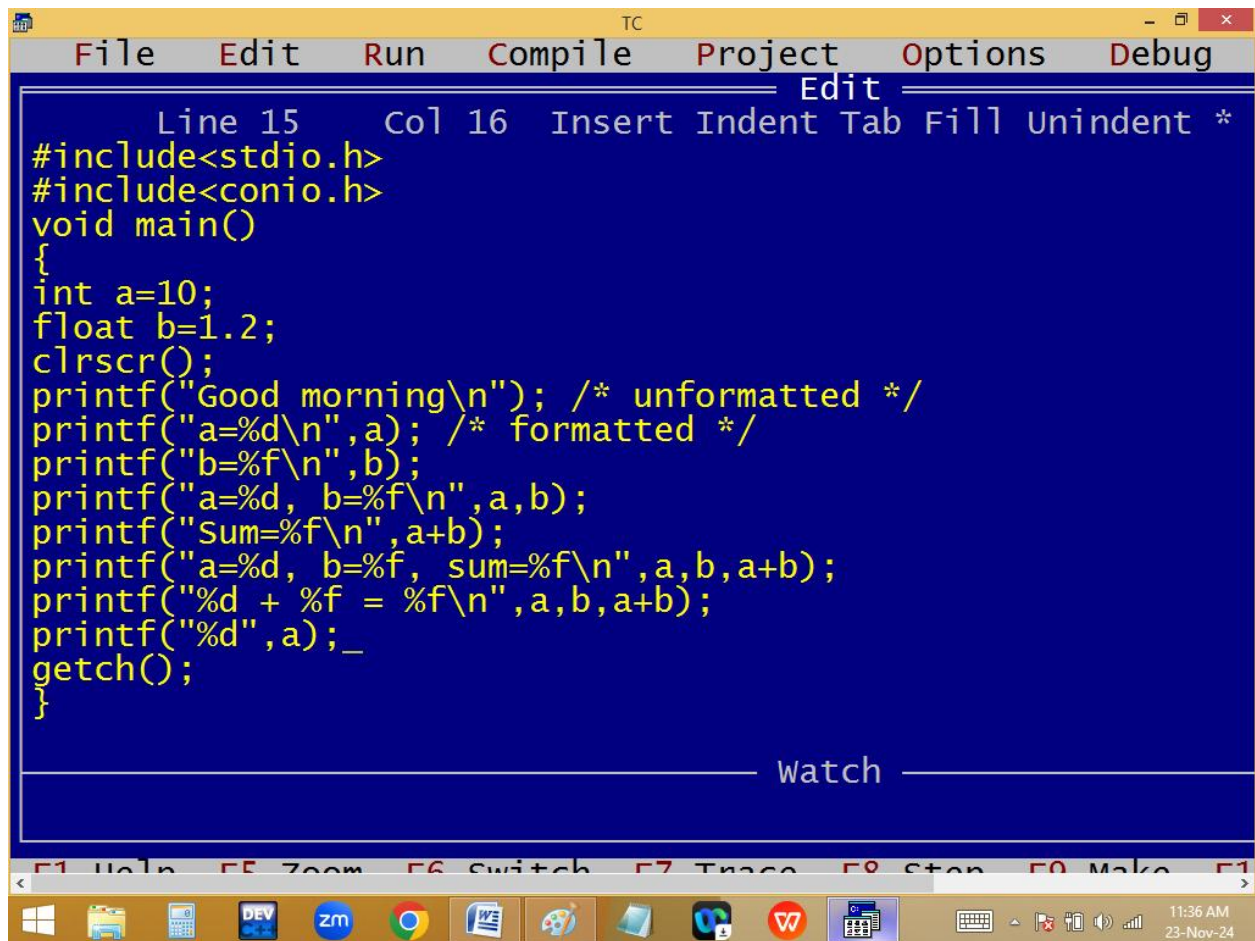


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The "Edit" menu is open, showing options: "Line 9", "Col 41", "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main text area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int c;
    clrscr();
    c = printf("Kishore");
    printf(" Length=%d\n", c);
    printf(" Length=%d", printf("Kishore"));_
    getch();
}
```

Below the code area is a "Watch" section, which is currently empty. At the bottom of the window, there is a status bar with function key shortcuts: "F1 Help", "F5 Zoom", "F6 Switch", "F7 Trace", "F8 Stop", "F9 Make", and "F10". The Windows taskbar is visible at the very bottom, showing various application icons and the system clock displaying "11:24 AM 23-Nov-24".

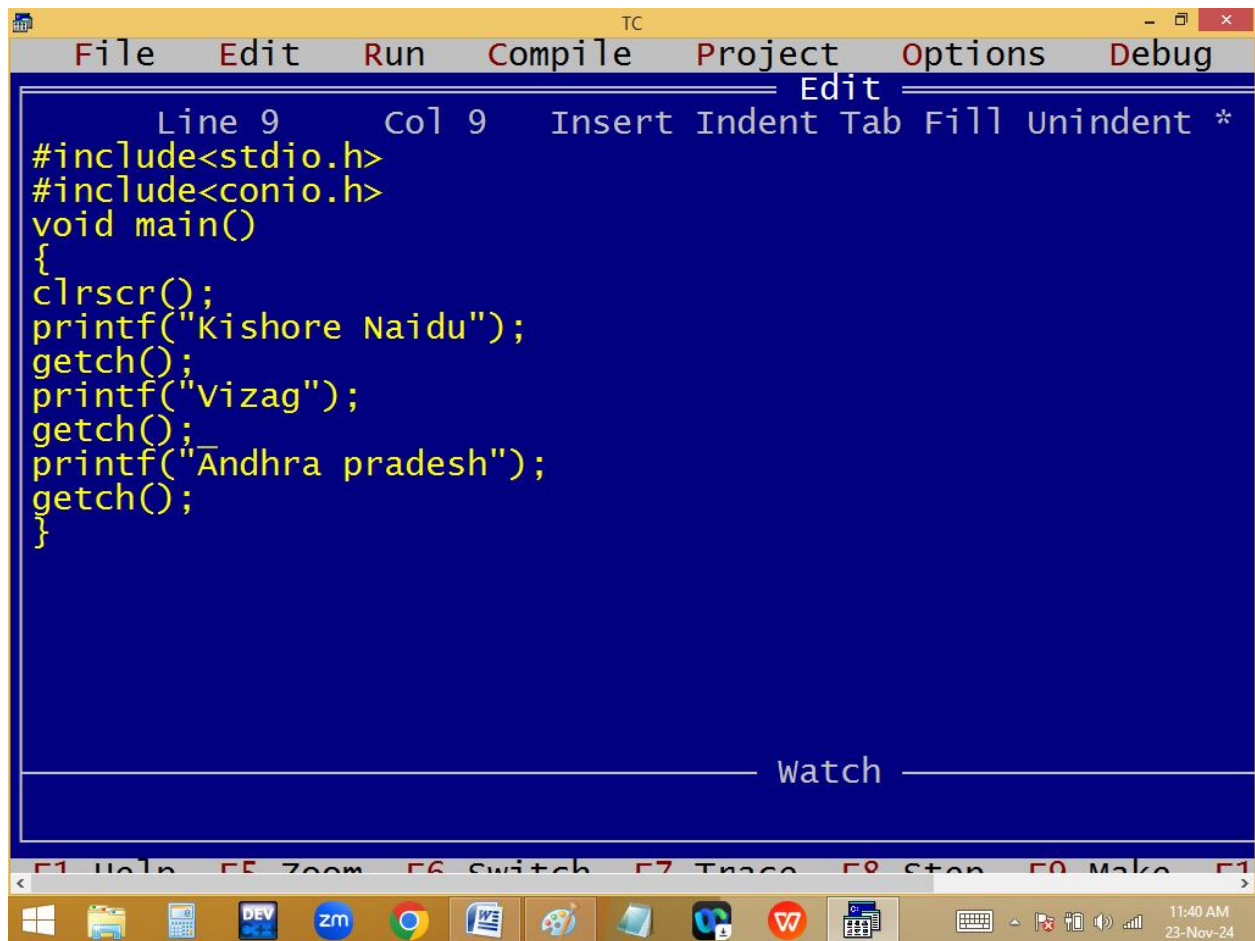




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The "Edit" menu is open, showing options: "Line 15", "Col 16", "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main text area has a dark blue background with yellow text. It contains a C program that declares two variables, a and b, and performs several printf statements to display their values and the sum. The program ends with a getch() call. Below the text area is a "Watch" window, which is currently empty. At the bottom of the screen is a Windows taskbar with various application icons and a system clock showing 11:36 AM on 23-Nov-24.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
float b=1.2;
clrscr();
printf("Good morning\n"); /* unformatted */
printf("a=%d\n",a); /* formatted */
printf("b=%f\n",b);
printf("a=%d, b=%f\n",a,b);
printf("Sum=%f\n",a+b);
printf("a=%d, b=%f, sum=%f\n",a,b,a+b);
printf("%d + %f = %f\n",a,b,a+b);
printf("%d",a);_
getch();
}
```

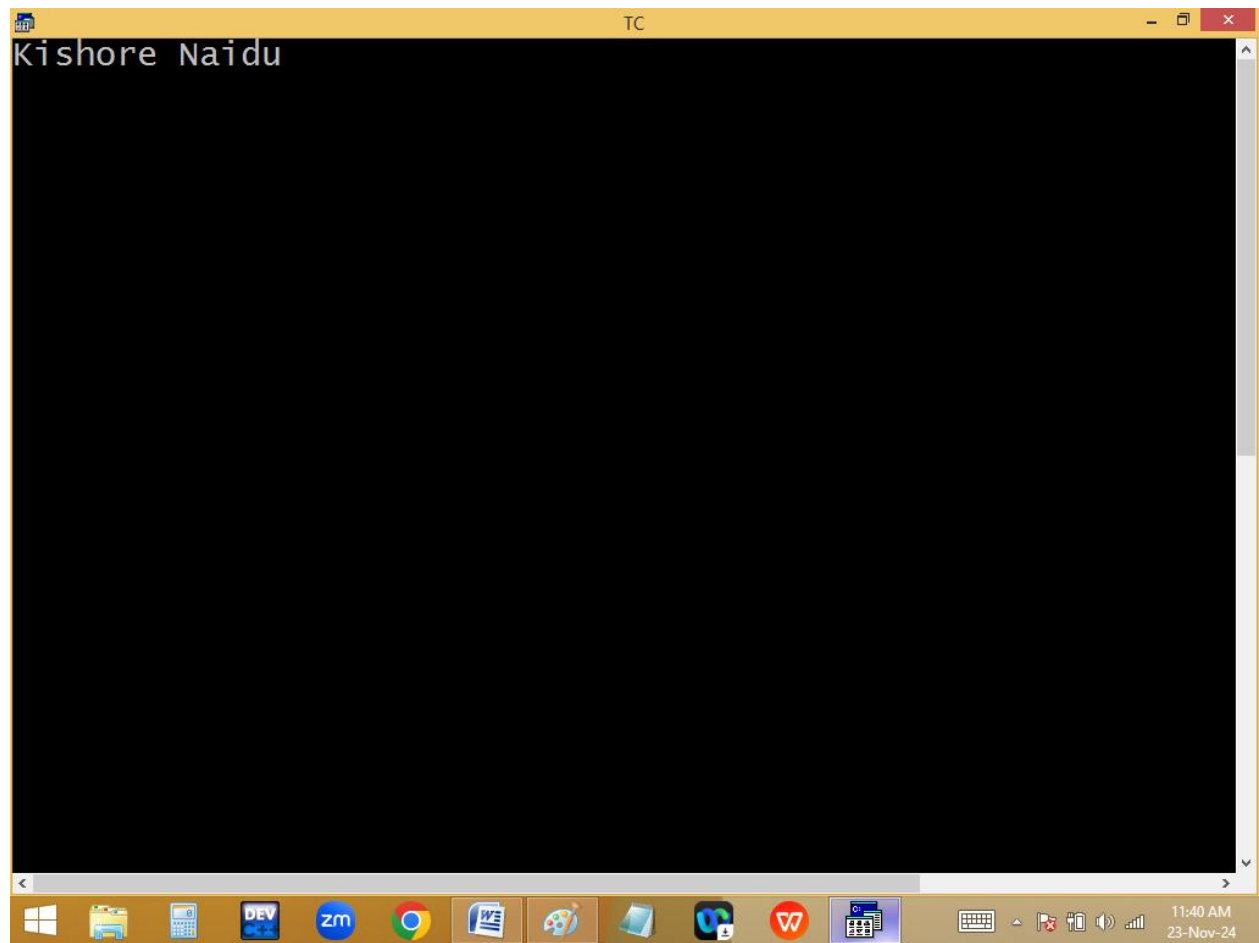
```
TC
Good morning
a=10
b=1.200000
a=10, b=1.200000
Sum=11.200000
a=10, b=1.200000, sum=11.200000
10 + 1.200000 = 11.200000
10_
```

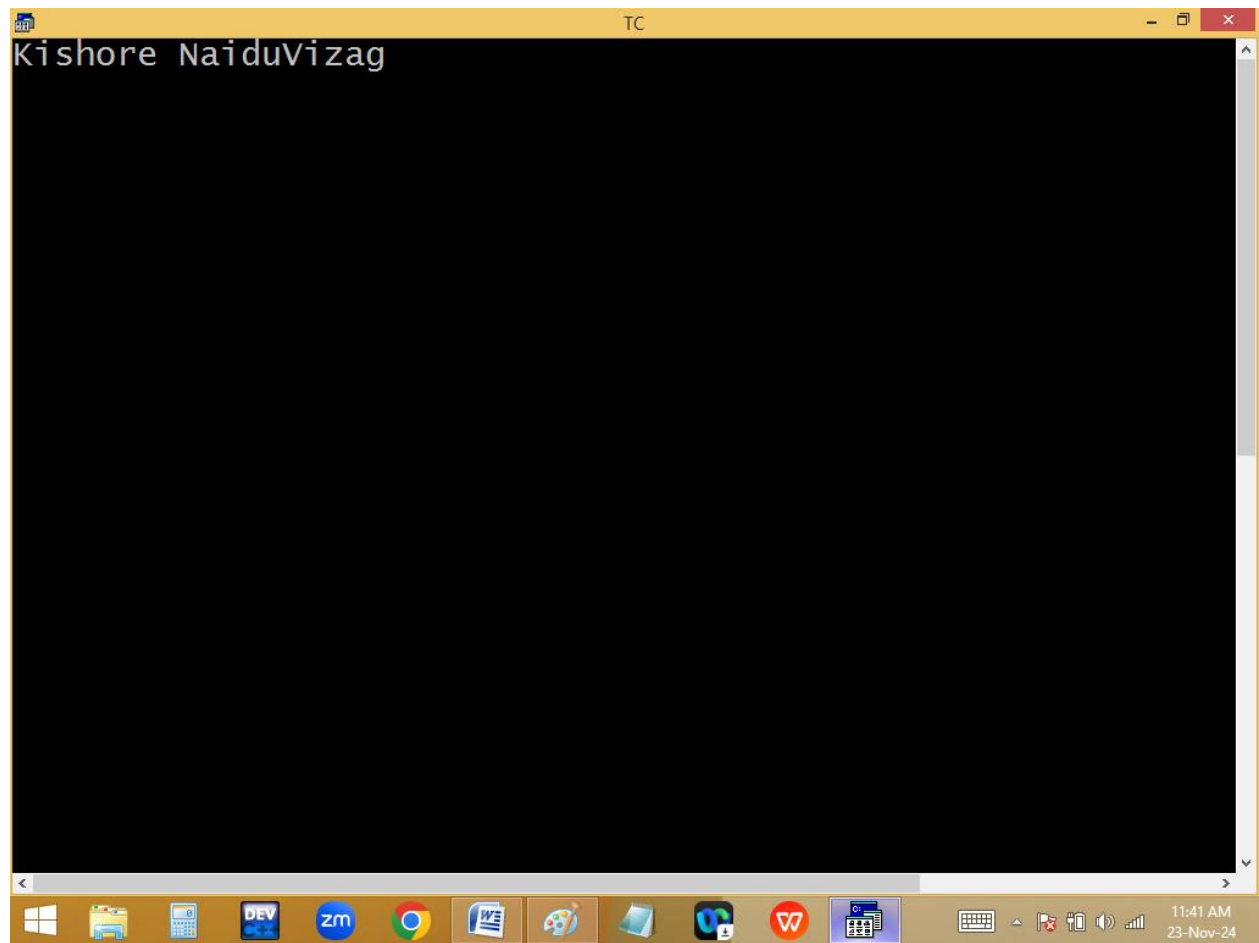


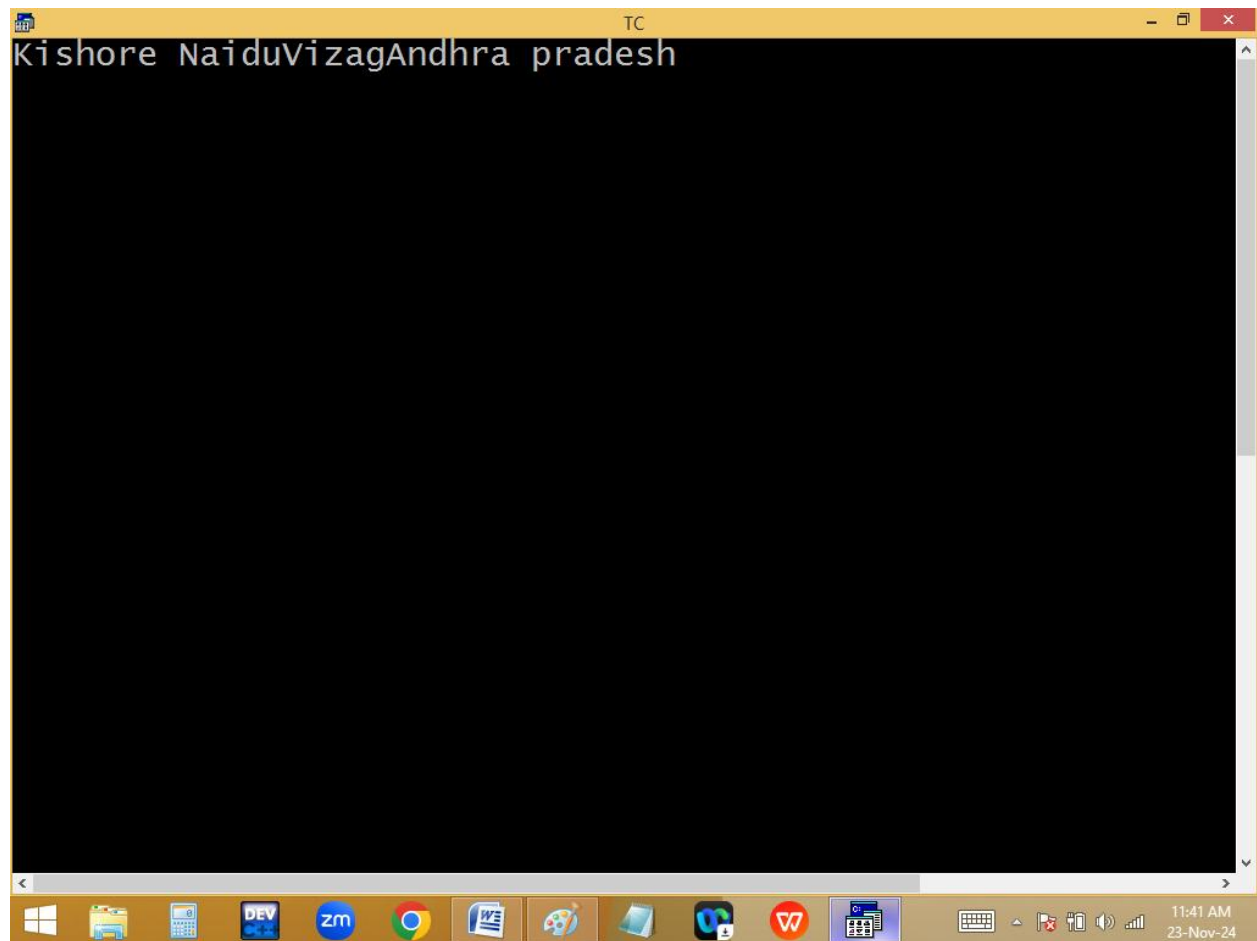
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The "Edit" menu is open, showing options: "Line 9", "Col 9", "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main text area has a dark blue background with yellow text. The code is as follows:

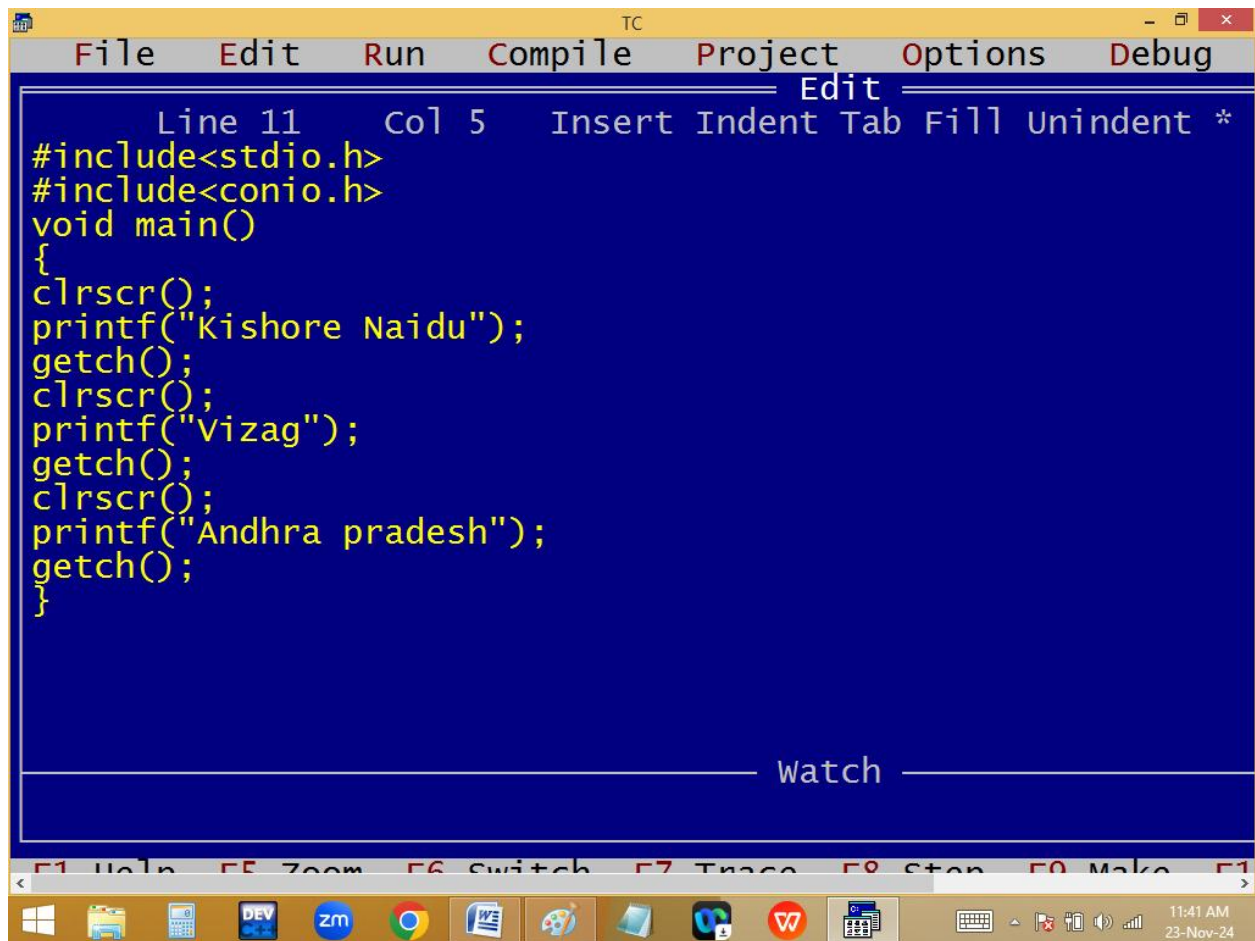
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Kishore Naidu");
getch();
printf("Vizag");
getch();
printf("Andhra pradesh");
getch();
}
```

Below the code area is a "Watch" section, which is currently empty. At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "11:40 AM" and date "23-Nov-24".





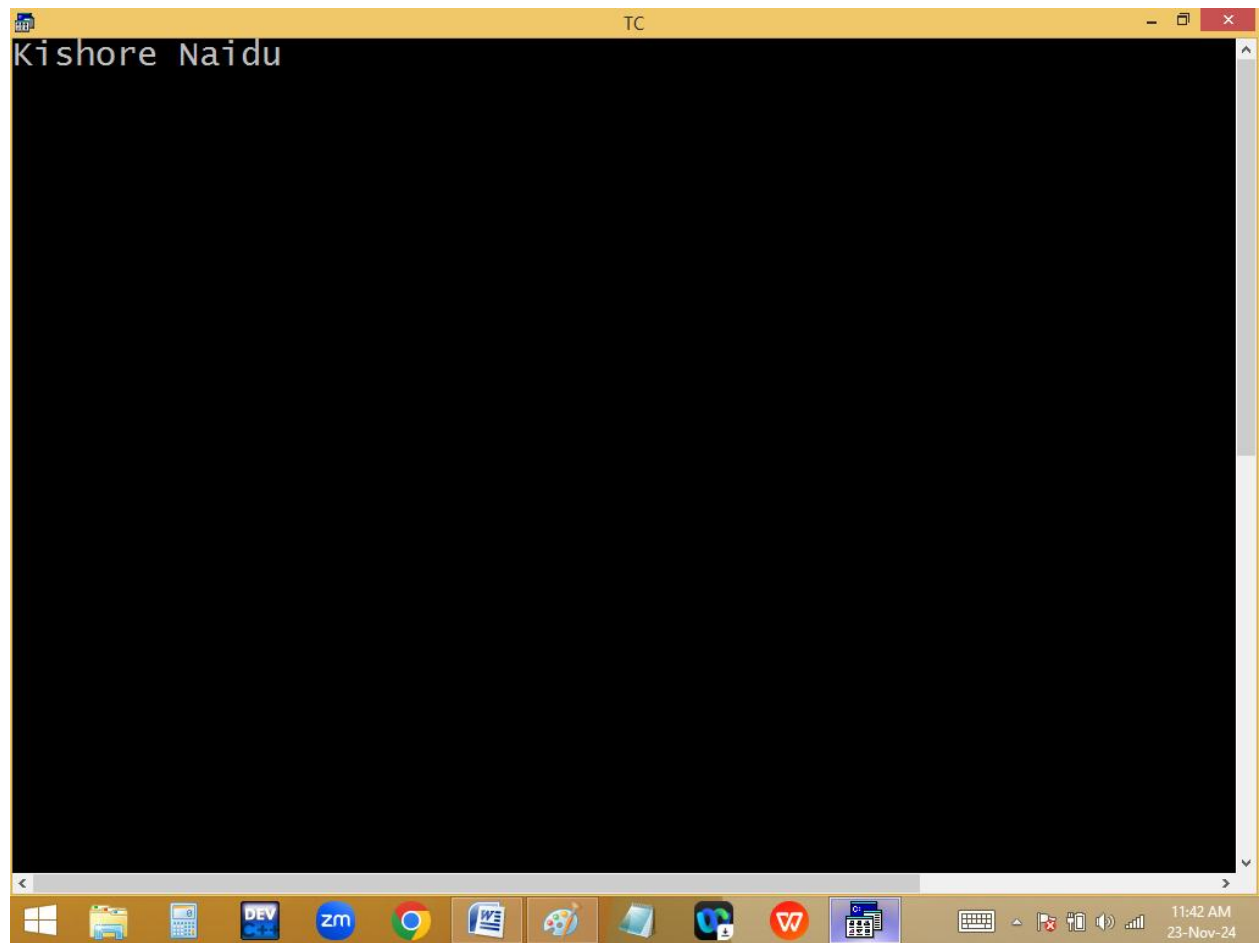


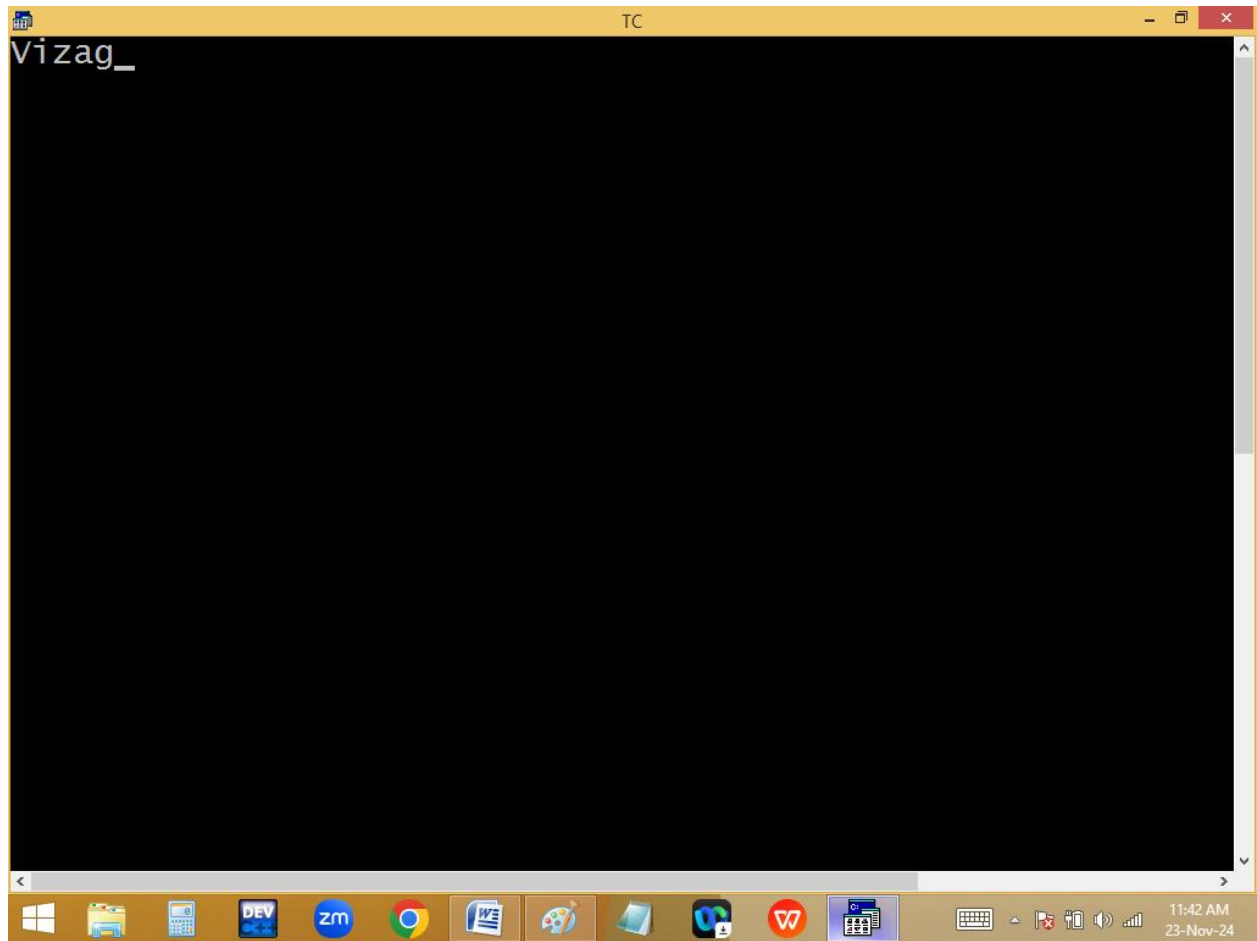


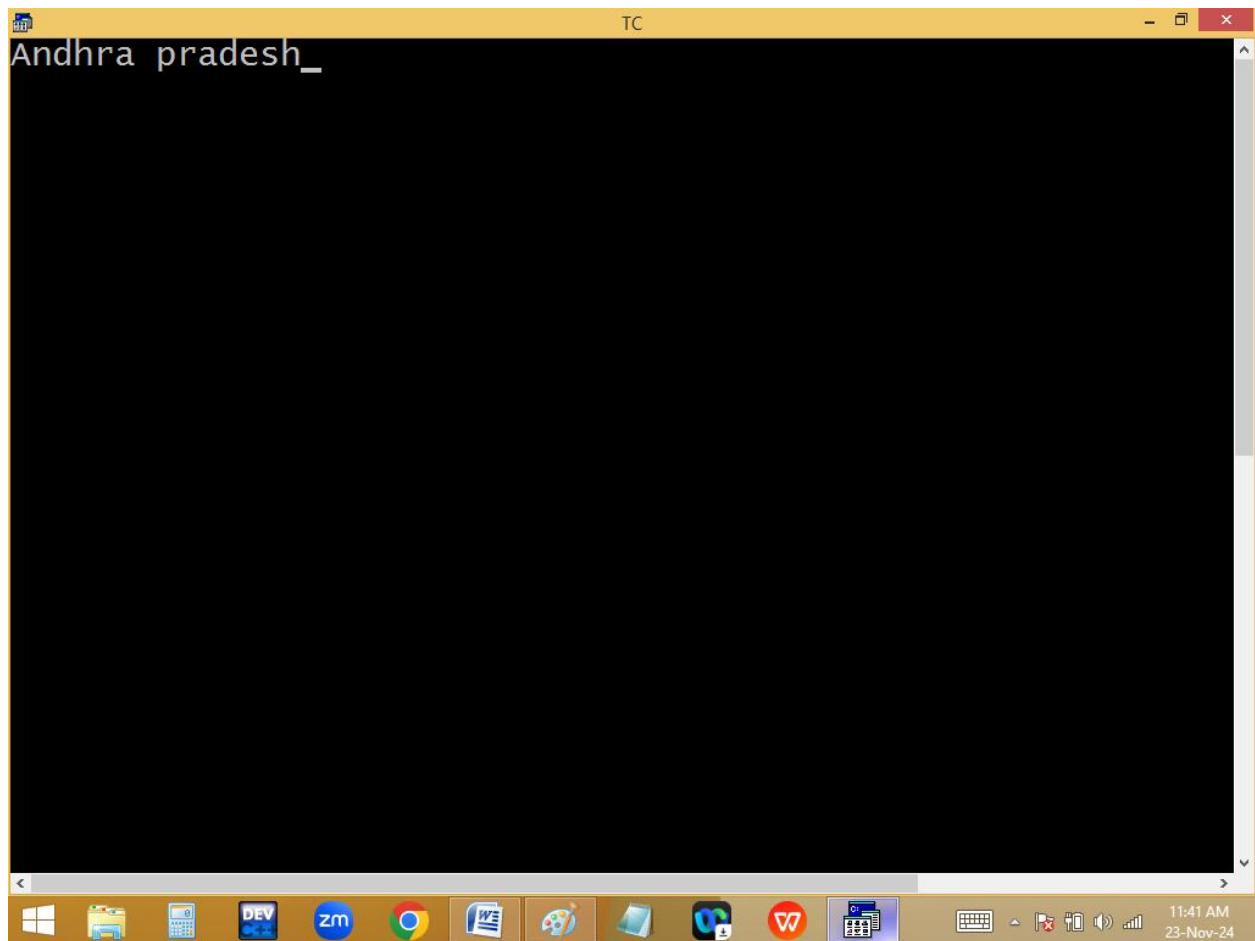
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The "Edit" menu is open, showing options: "Line 11", "Col 5", "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main text area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Kishore Naidu");
getch();
clrscr();
printf("Vizag");
getch();
clrscr();
printf("Andhra pradesh");
getch();
}
```

Below the code editor is a "Watch" window, which is currently empty. At the bottom of the IDE, there is a toolbar with icons for various functions, and a status bar showing the time "11:41 AM" and date "23-Nov-24".







```
#include<stdio.h>

#include<conio.h>

#include<stdlib.h>

int main()

{

    system("cls");

    printf("Hello");

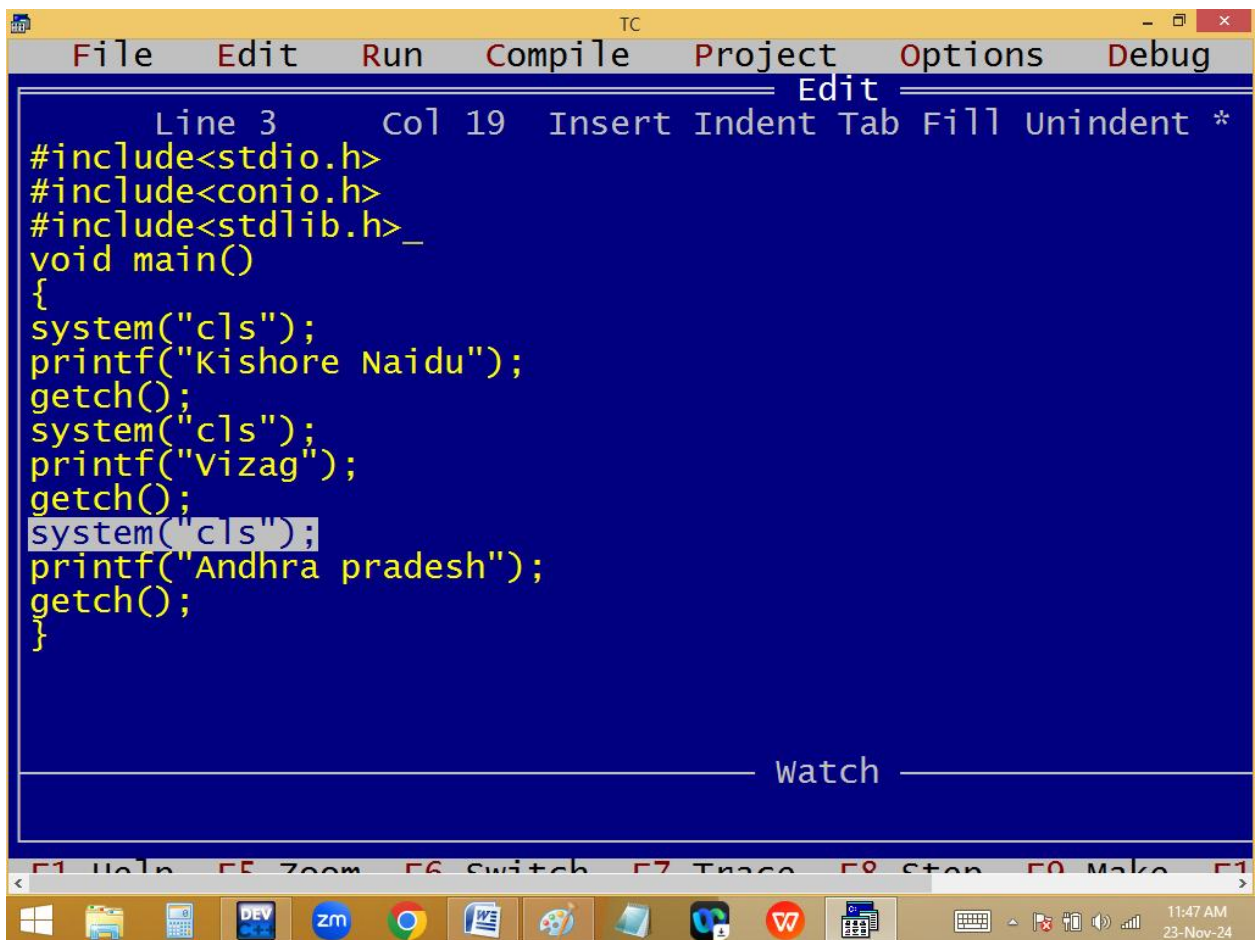
    getch();
```

```
    system("cls");  
    printf("Thank you");  
    getch();  
}
```

//f9 - compile

//f10 - run

//f11 - compile and run



```
TC  
File Edit Run Compile Project Options Debug  
Edit  
Line 3 Col 19 Insert Indent Tab Fill Unindent *  
#include<stdio.h>  
#include<conio.h>  
#include<stdlib.h>_  
void main()  
{  
    system("cls");  
    printf("Kishore Naidu");  
    getch();  
    system("cls");  
    printf("Vizag");  
    getch();  
    system("cls");  
    printf("Andhra pradesh");  
    getch();  
}
```

Watch

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

11:47 AM
23-Nov-24

BACK SLASH / ESCAPE SEQUENCE CHARACTERS

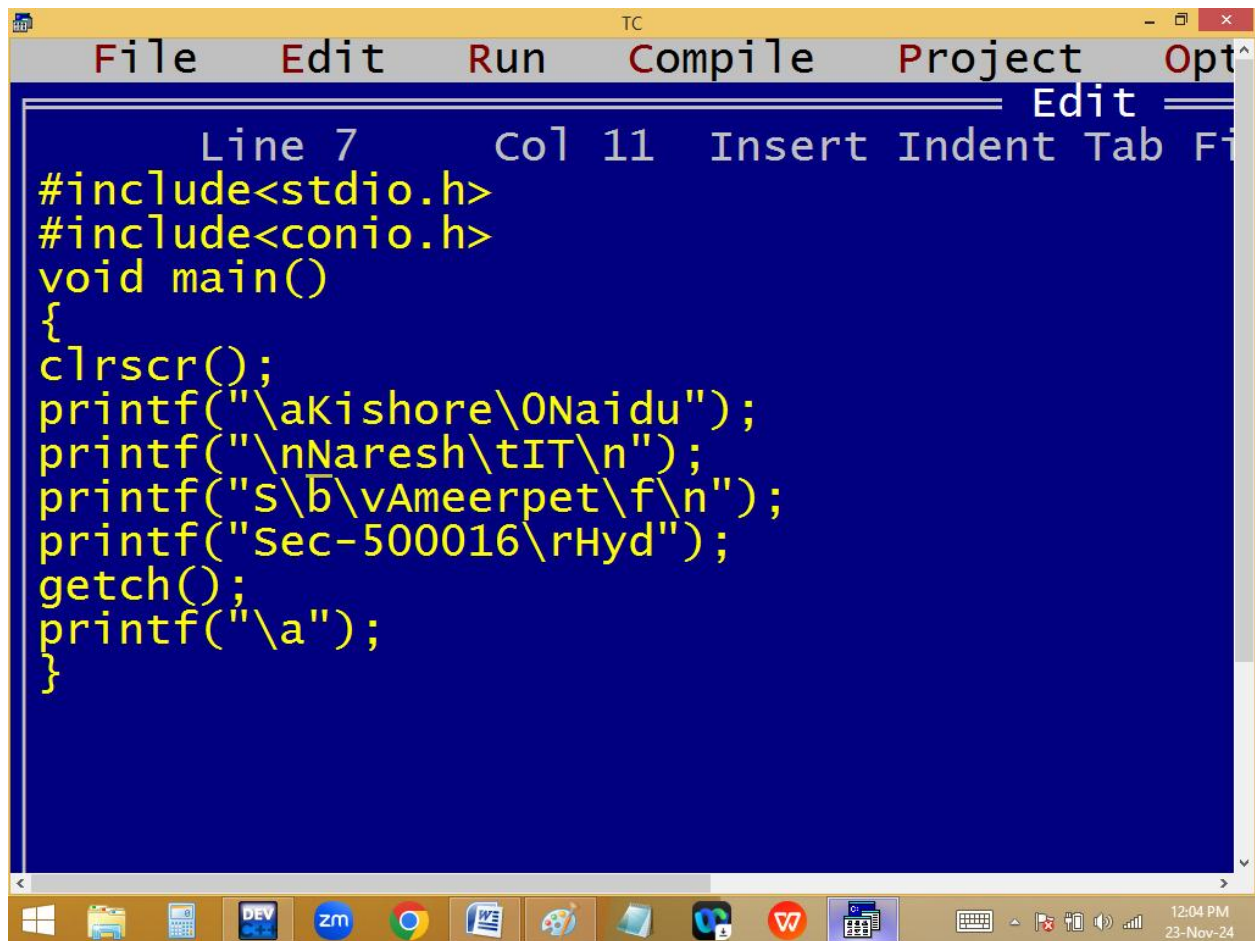
They started with back slash [\].

They used to format the outputs.

They participated in program execution but not displayed in output. Hence they are also called **escape sequence characters**.

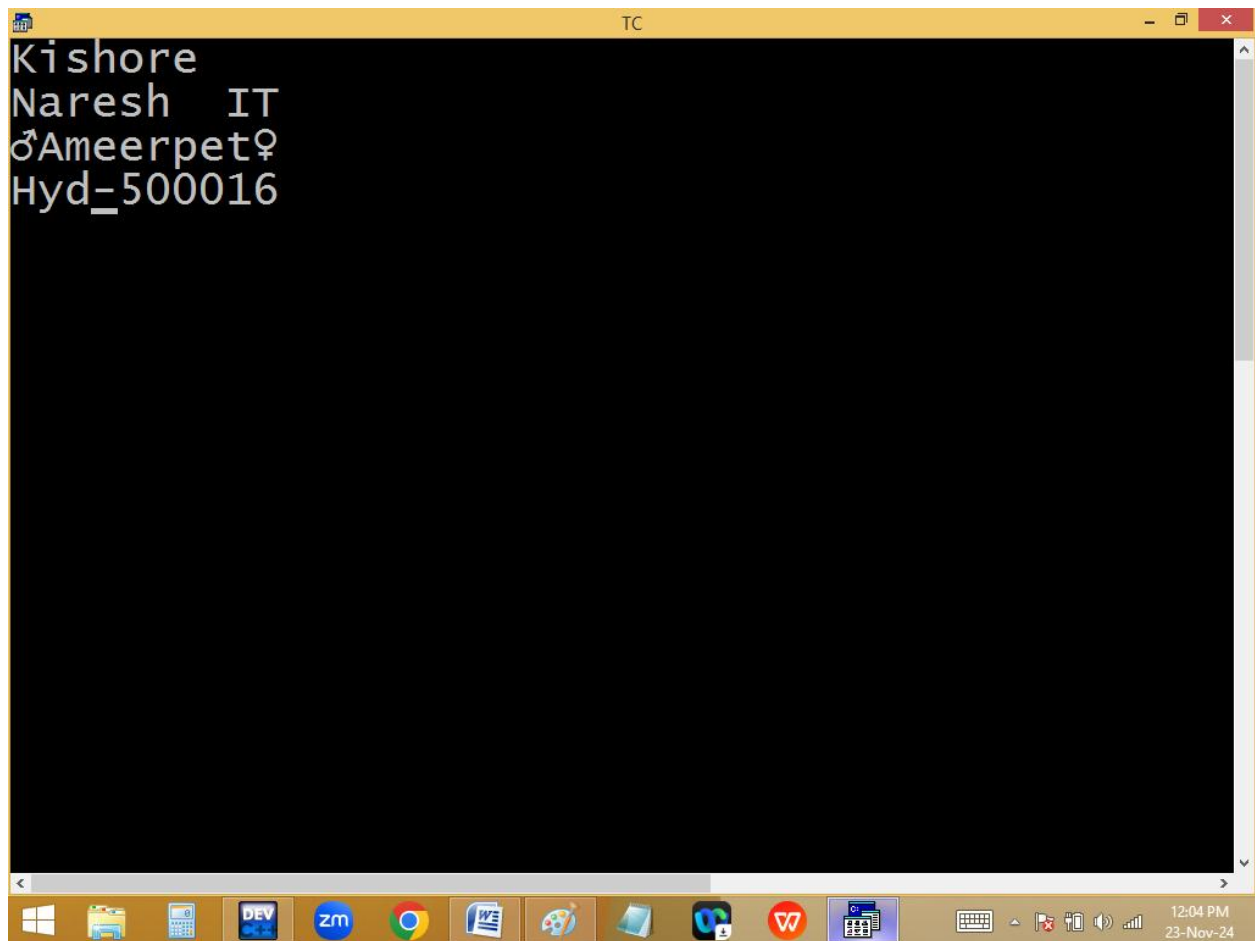
Each back slash character=**1 byte i.e. one character**.

BACK SLASH CHARACTER	DESCRIPTION
\a	Alert [beep sound]
\b	Back space
\n	New line character
\t	Tab space
\r	Carriage return[beginning of line]
\f	Form feed ♀
\v	Vertical tab ♂
\0	Null char
\\	\ [invalid]
\k	k [invalid]

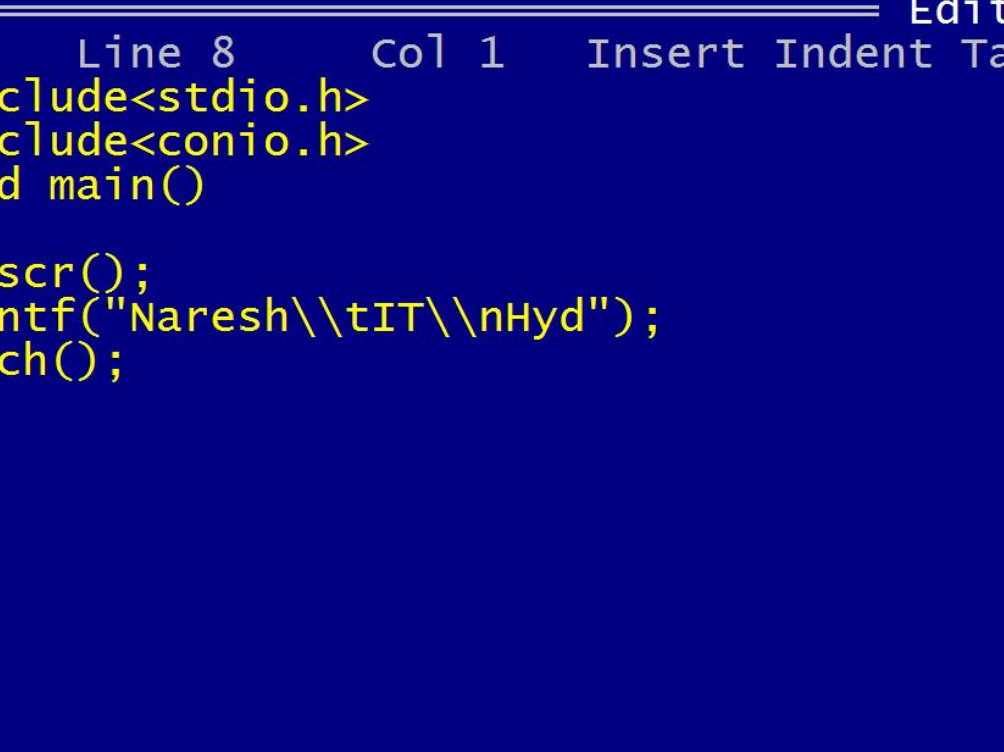


```
Line 7      Col 11  Insert  Indent  Tab  Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("\akishore\0Naidu");
printf("\nNareshtIT\n");
printf("S\b\vAmeerpet\f\n");
printf("Sec-500016\rHyd");
getch();
printf("\a");
}
```

```
TC
Kishore
Naresh IT
♂Ameerpet♀
Hyd_500016
```



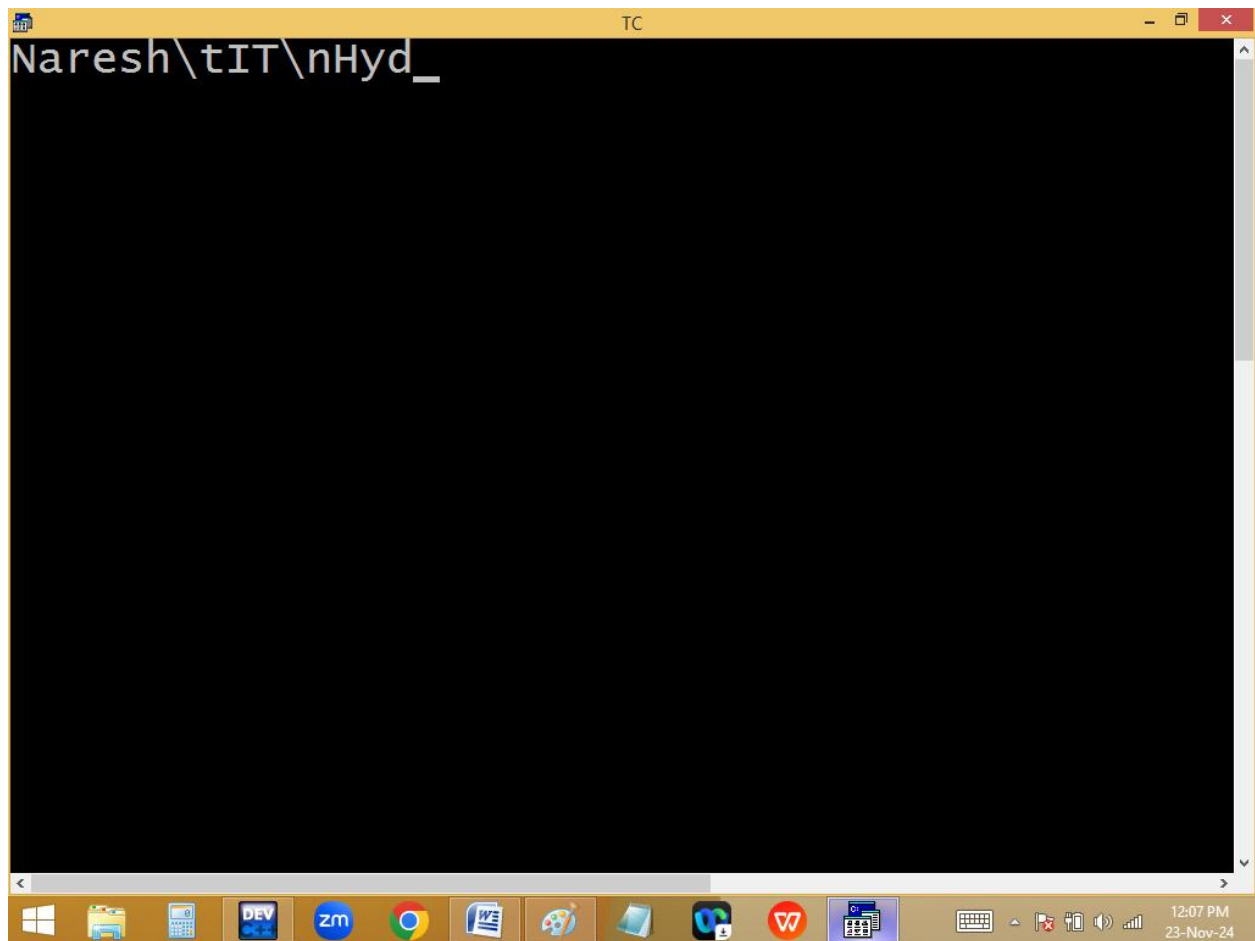
Printing back slash characters:



The screenshot shows the Turbo C++ IDE interface. The menu bar at the top includes File, Edit, Run, Compile, Project, and Options. The status bar at the bottom indicates Line 8, Column 1. The main editing area contains the following C code:

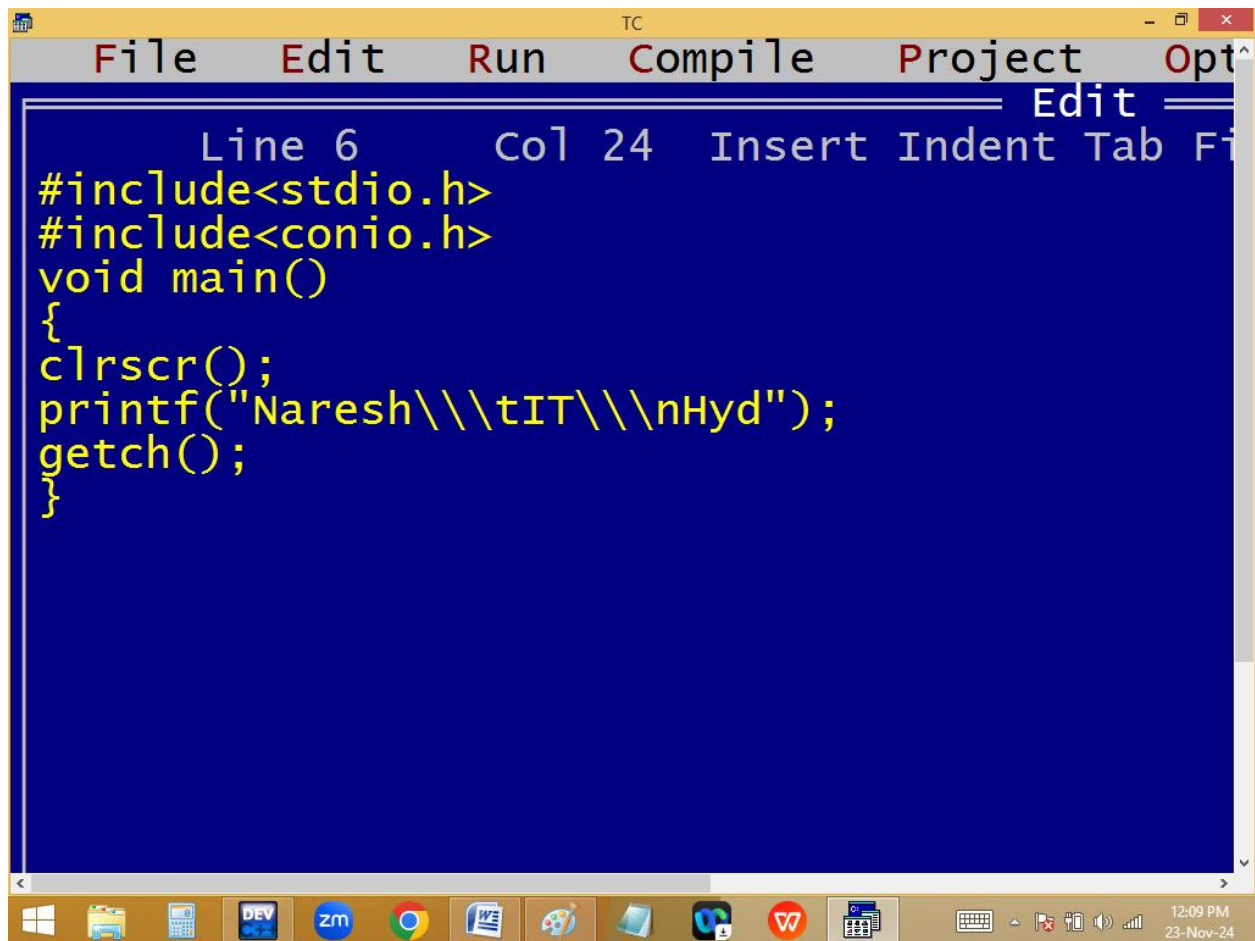
```
#include<stdio.h>
#include<conio.h>
void main()
{
    clrscr();
    printf("Naresh\\tIT\\nHyd");
    getch();
}
```

The code is written in yellow text on a blue background. The output of the program, "Naresh\\tIT\\nHyd", is visible in the output window at the bottom right, which also shows the execution time as 0.000 seconds.



p("Naresh\\tIT\\nHyd");

Naresh\tIT\nHyd



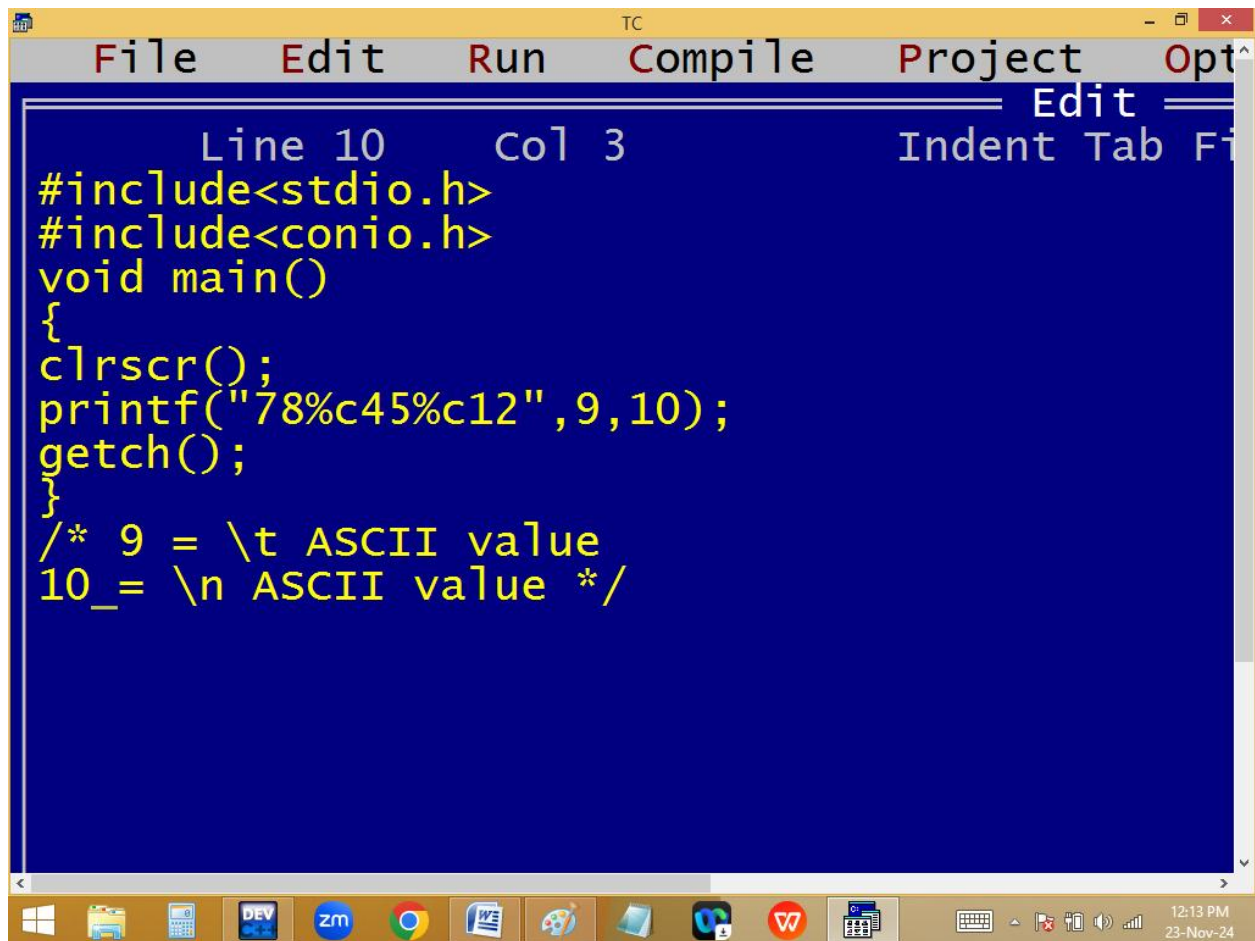
```
Line 6      col 24  Insert Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Naresh\\tIT\\nHyd");
getch();
}
```

12:09 PM
23-Nov-24

```
Naresh\      IT\  
Hyd_
```

p("Naresh\\tIT\\nHyd");

Naresh\ IT\
Hyd

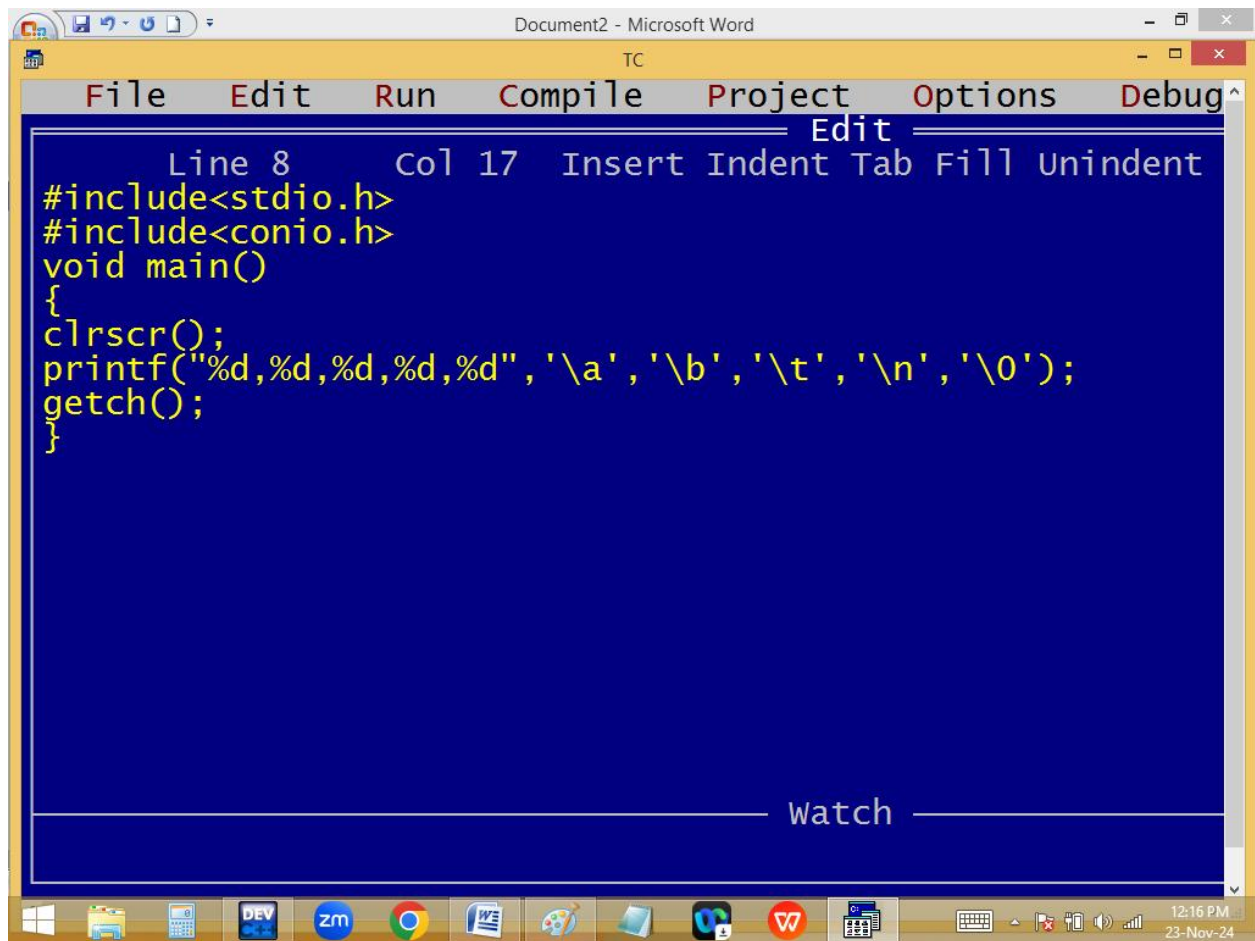


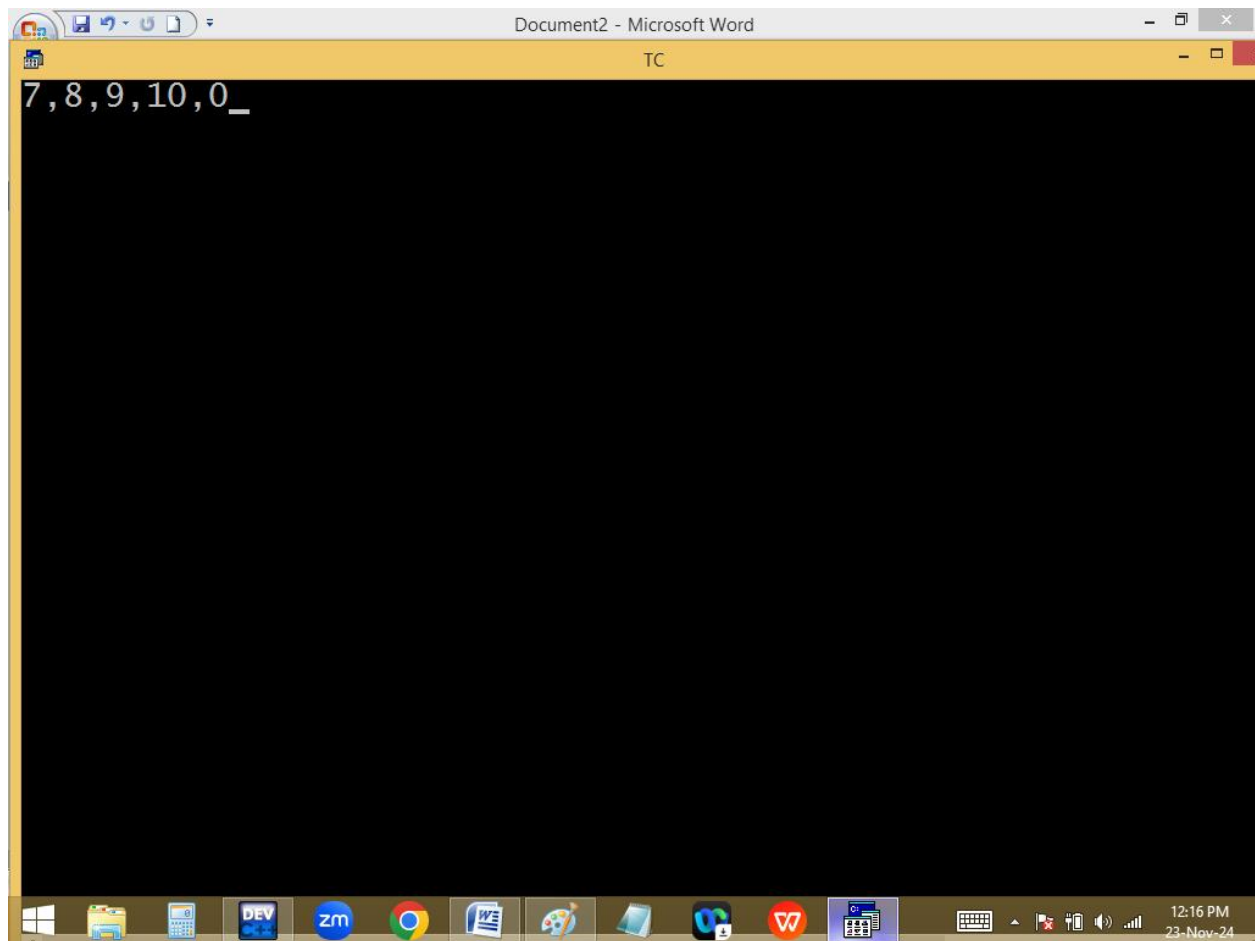
```
File Edit Run Compile Project Options
Line 10 col 3 Indent Tab Fi
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("78%c45%c12",9,10);
getch();
}
/* 9 = \t ASCII value
10_ = \n ASCII value */
```

12:13 PM
23-Nov-24

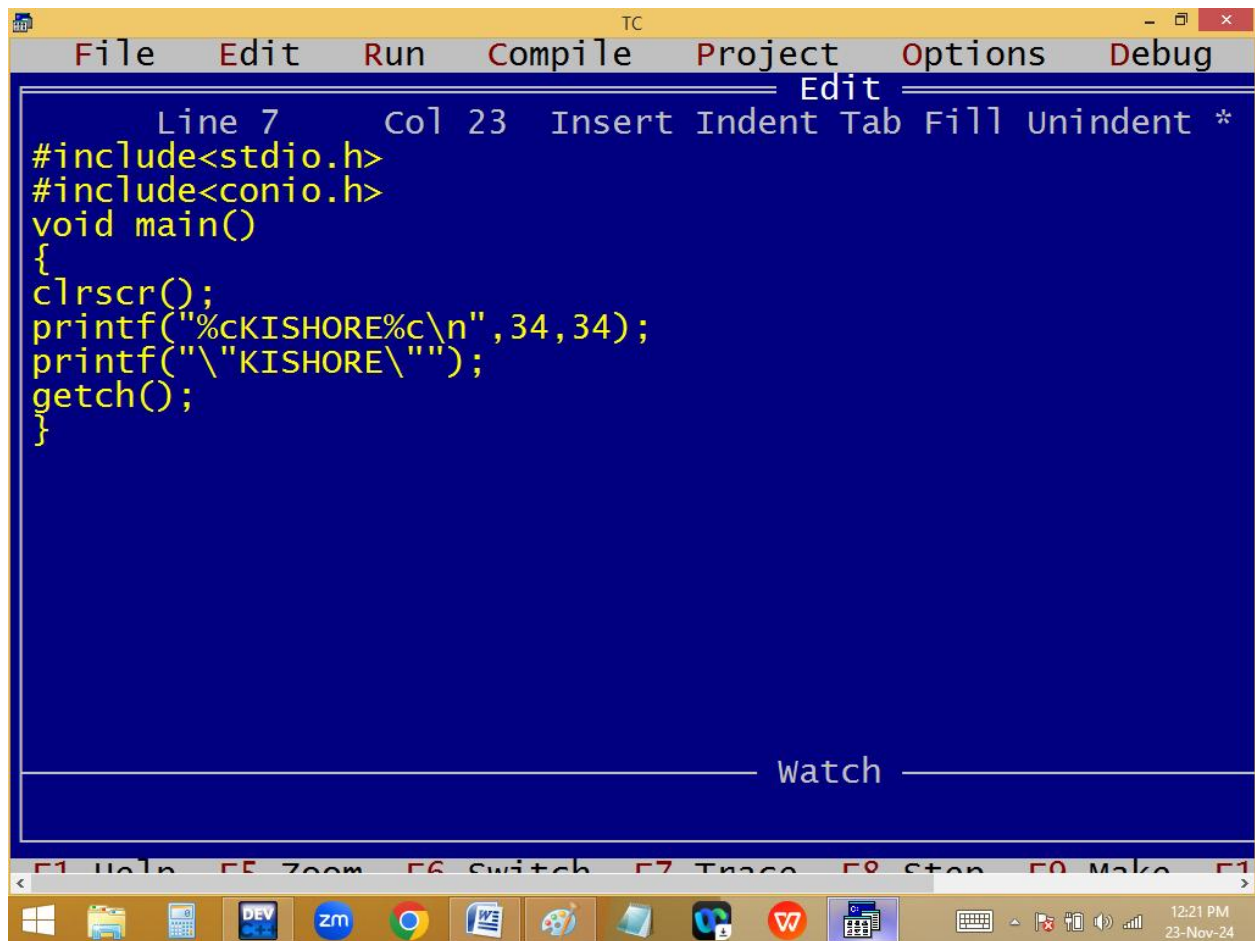
```
TC
78      45
12
```

`\t \n`
`p( "78%c45%c12", 9, 10 );`
`\t \n`





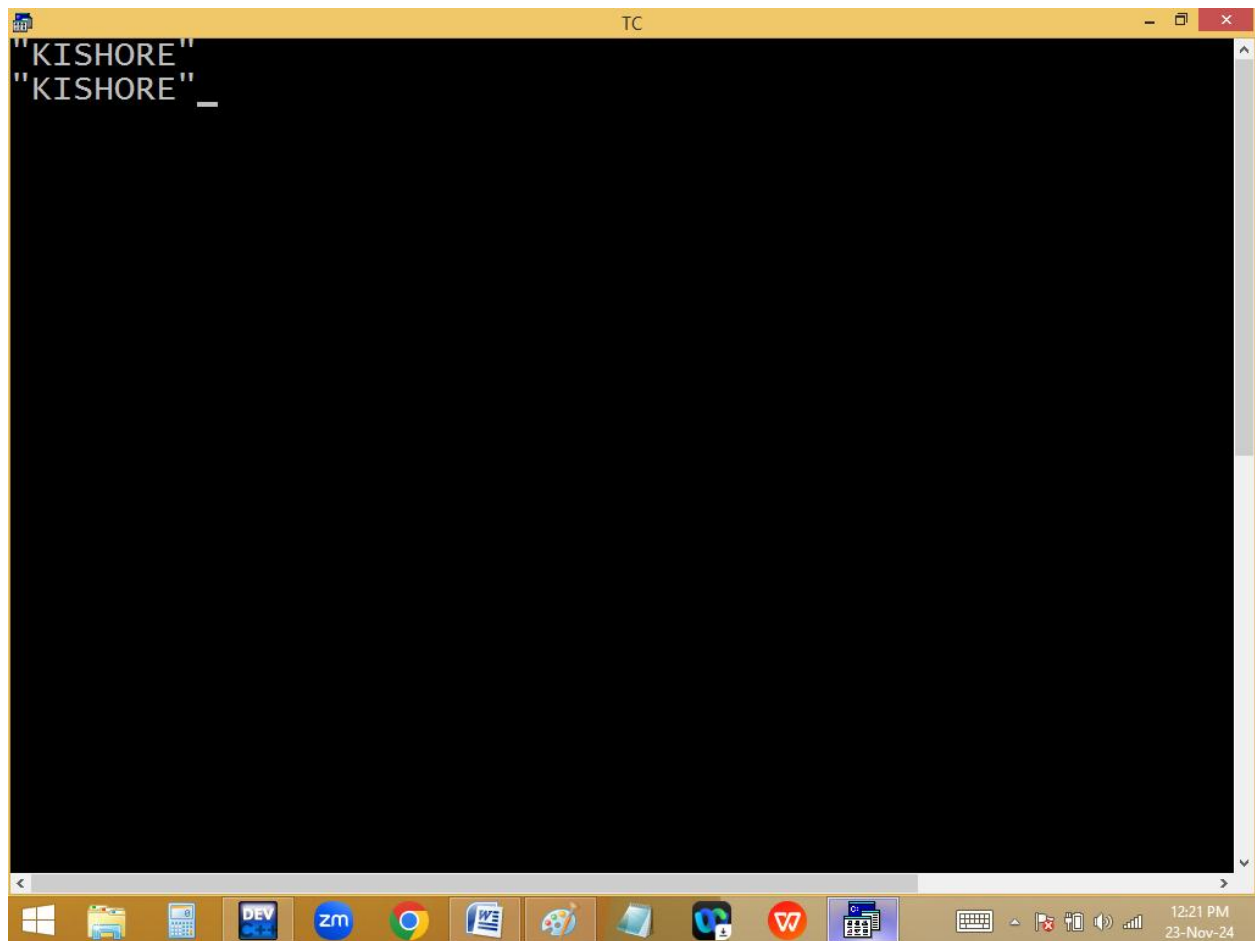
Printing name in “ “



The image shows a screenshot of the Turbo C++ (TC) IDE. The main window has a dark blue background with yellow text. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, and Debug. Below the menu bar, a status bar shows 'Line 7', 'Col 23', and various editing options like Insert, Indent, Tab, Fill, and Unindent. The code being edited is a C program that prints 'KISHORE' in a box of 34x34 characters. The code is as follows:

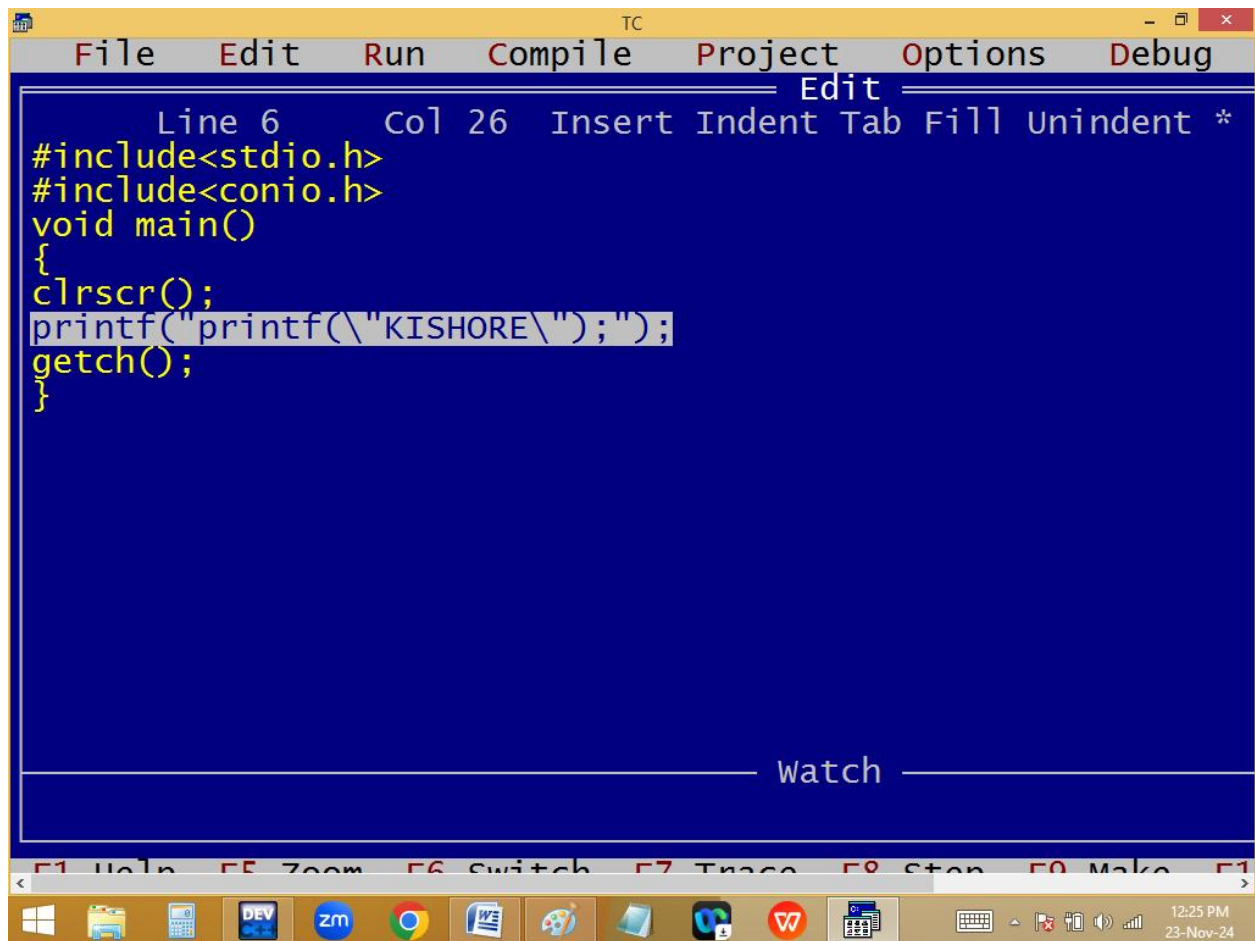
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%cKISHORE%c\n",34,34);
printf("\nKISHORE\n");
getch();
}
```

At the bottom of the IDE window, there is a 'Watch' section which is currently empty. Below the IDE window, the Windows taskbar is visible, showing various application icons and the system clock indicating 12:21 PM on 23-Nov-24.



The image shows a screenshot of a Turbo C++ (TC) compiler window. The window has a yellow title bar with the text "TC" and standard Windows window controls (minimize, maximize, close). The main area of the window is black, representing the output console. In the top-left corner of this black area, the text `"KISHORE"` is displayed on the first line, and `"KISHORE"` followed by a cursor underline (`_`) is displayed on the second line. At the bottom of the window is a Windows taskbar with various application icons including File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and others. The system clock in the bottom right corner shows the time as 12:21 PM on 23-Nov-24.

```
printf("\\"KISHORE\\");
```



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The "Edit" menu is open, showing options: "Line 6", "Col 26", "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The code editor has a dark blue background and contains the following C code:

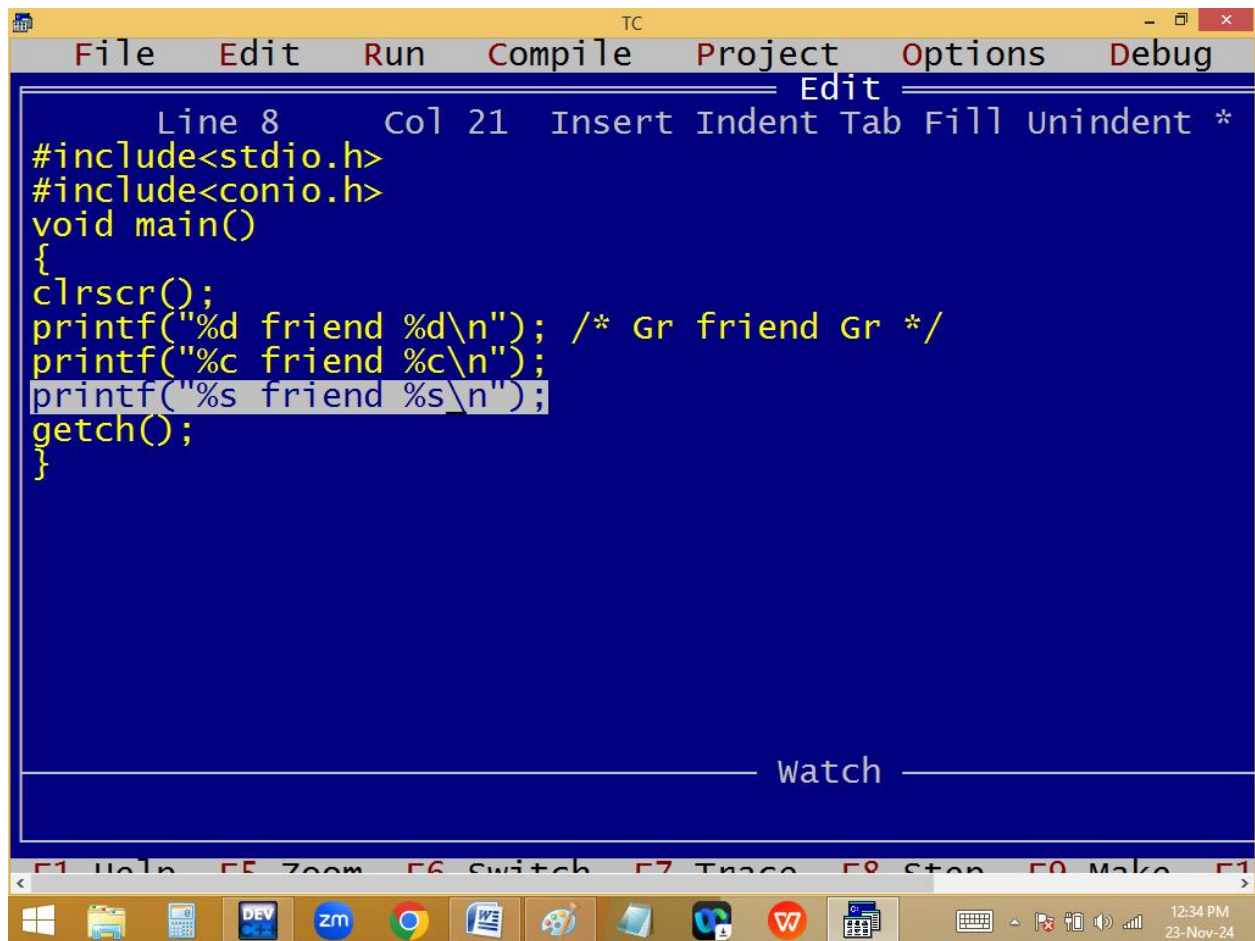
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("printf(\"KISHORE\");");
getch();
}
```

Below the code editor is a "Watch" window, which is currently empty. At the bottom of the IDE, there is a toolbar with icons for various functions, and a status bar showing the time "12:25 PM" and date "23-Nov-24".

```
TC
printf("KISHORE");_
```

TEXT

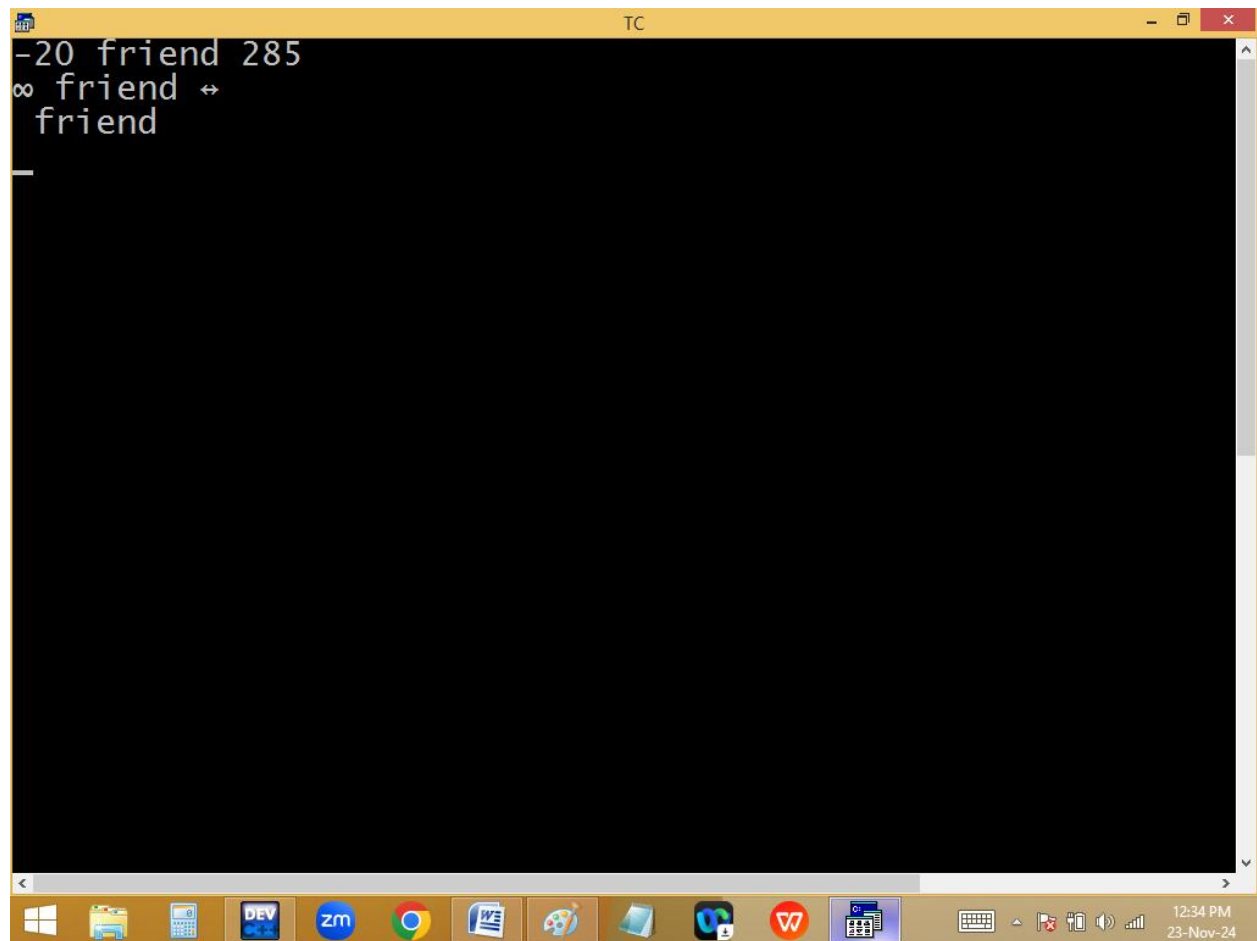
```
printf("printf(\"KISHORE\");");
```

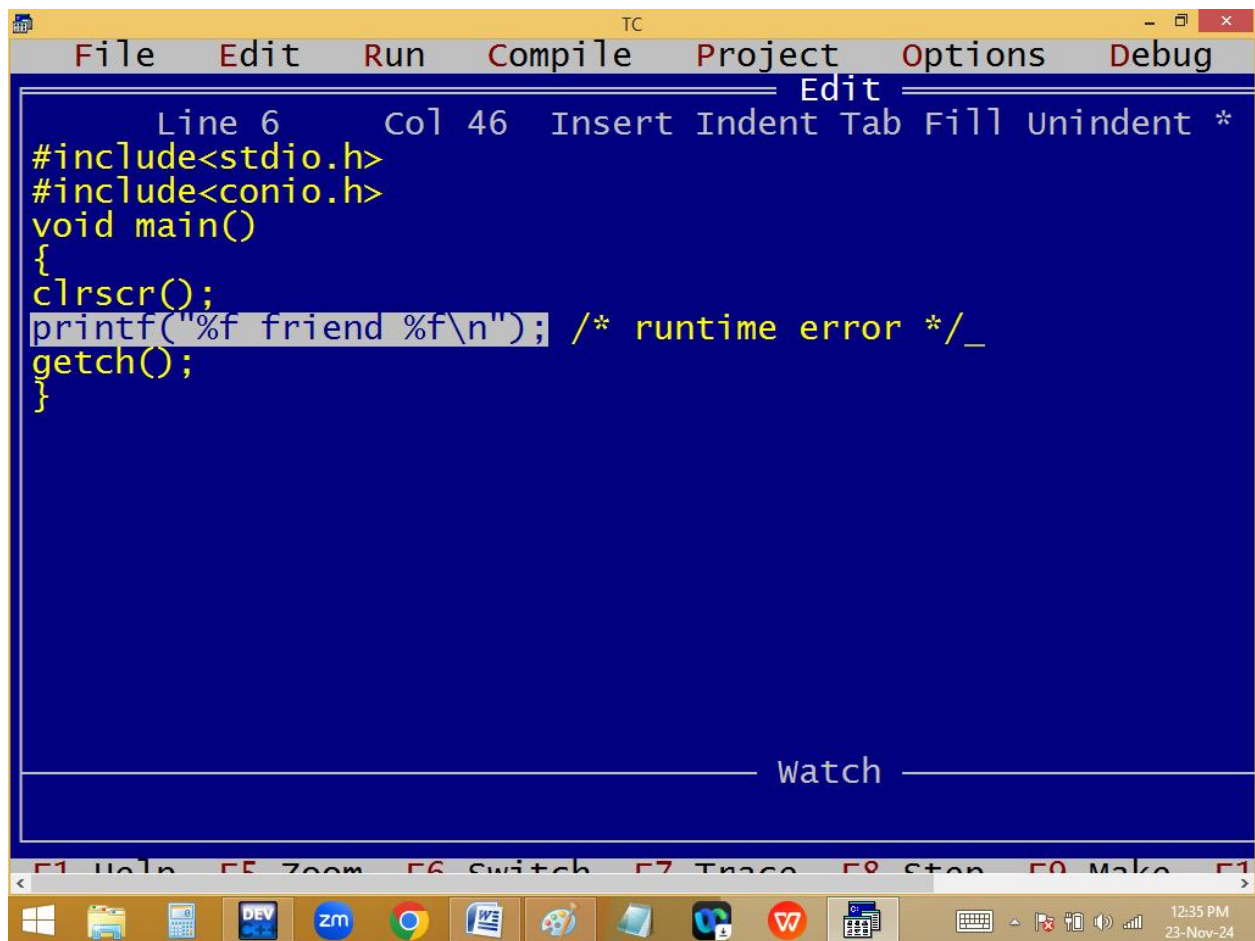


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 8", "Col 21", and lists editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. The code is as follows:

```
Line 8      Col 21  Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d friend %d\n"); /* Gr friend Gr */
printf("%c friend %c\n");
printf("%s friend %s\n");
getch();
}
```

Below the code editor is a "Watch" window, which is currently empty. At the bottom of the screen is a Windows taskbar with various application icons (Windows Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, VS Code, and a red 'W' icon). The system clock in the bottom right corner shows "12:34 PM" and "23-Nov-24".

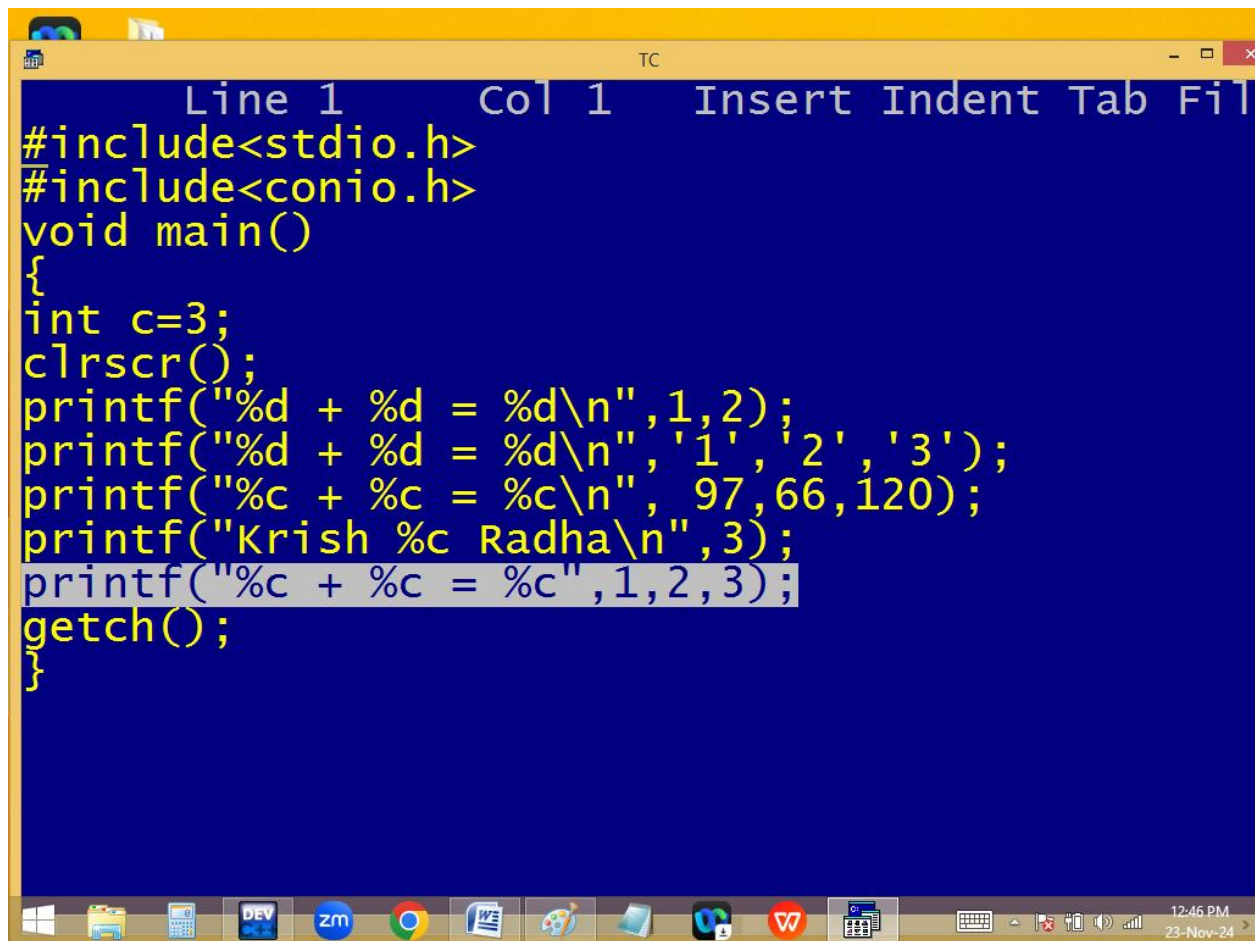




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The "Edit" menu is open, showing options: "Line 6", "Col 46", "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The code editor has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f friend %f\n"); /* runtime error */_
getch();
}
```

The line containing the printf statement is highlighted in light blue. Below the code editor is a "Watch" window, which is currently empty. At the bottom of the IDE, there is a toolbar with icons for various functions, and a status bar showing the time "12:35 PM" and date "23-Nov-24".



```
Line 1    Col 1    Insert Indent Tab Fil
#include<stdio.h>
#include<conio.h>
void main()
{
int c=3;
clrscr();
printf("%d + %d = %d\n",1,2);
printf("%d + %d = %d\n",'1','2','3');
printf("%c + %c = %c\n", 97,66,120);
printf("Krish %c Radha\n",3);
printf("%c + %c = %c",1,2,3);
getch();
}
```

